



**NATIONAL UNIVERSITY**  
of Computer & Emerging Sciences

## **Software Engineering Project Report**

Retail-IQ – Smart Retail Demand Forecasting System

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Submitted to:

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Date of submission: 3/5/26

# **Smart Retail Demand Forecasting System**

## **(RetailIQ)**

### **Project Overview**

RetailIQ is a web-based retail analytics platform designed to help businesses forecast product demand and understand the factors influencing sales performance. The system uses historical retail sales data to generate demand forecasts and provide meaningful insights through interactive dashboards and analytical tools.

Traditional retail forecasting tools only predict future sales numbers but fail to explain the reasons behind demand fluctuations. RetailIQ improves this by providing interpretable analysis of sales patterns, promotion impacts, and product relationships. The platform allows users to upload sales datasets, visualize sales trends, analyze promotional effects, and receive business recommendations based on data-driven insights.

The system integrates machine learning models with a user-friendly web interface to help retail managers make informed decisions regarding inventory management, promotions, and product stocking strategies. Through interactive visualizations and analytical dashboards, RetailIQ transforms complex retail data into actionable insights.

# User Stories

Sprint 1 focuses on data preparation and exploratory analysis. Advanced forecasting and prediction features will be implemented in future sprints

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| <b>Story ID:</b> US-01Story<br><b>Title:</b> Data Loading                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Priority:</b> High<br><b>Estimated Hours:</b> 6 |
| <b>User Story:</b><br><br>As a retail analyst, I want to upload and access retail sales data in the system so that I can analyze sales performance easily.<br><br><b>Description:</b><br><br>This story involves enabling users to upload various retail datasets (such as sales, stores, and oil data), validate them, and provide a summary view for quick understanding.<br><br><b>Sub User Stories:</b> <ul style="list-style-type: none"><li>• As a user, I want to upload datasets (sales, stores, oil).</li><li>• As a user, I want the system to validate uploaded data.</li><li>• As a user, I want to view dataset summaries.</li></ul><br><b>Acceptance Criteria:</b> <ul style="list-style-type: none"><li>• Users should be able to upload datasets successfully.</li><li>• The system should validate the data and show errors if any.</li><li>• A summary view of uploaded data should be displayed.</li></ul> |                                                    |

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| <b>Story ID: US-02</b><br><b>Story Title: Data Cleaning</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Priority: High</b><br><b>Estimated Hours: 7</b> |
| <b>User Story:</b><br><p>As a retail analyst, I want the system to automatically clean and preprocess data so that I can work with accurate and reliable information.</p> <b>Description:</b><br><p>This story focuses on automating data preprocessing tasks such as handling missing values, detecting outliers, and identifying holidays that impact sales.</p> <b>Sub User Stories:</b> <ul style="list-style-type: none"> <li>• As a user, I want missing values to be handled automatically.</li> <li>• As a user, I want unusual sales values (outliers) to be detected.</li> <li>• As a user, I want holidays affecting sales to be identified.</li> </ul> <b>Acceptance Criteria:</b> <ul style="list-style-type: none"> <li>• Missing values should be handled appropriately.</li> <li>• Outliers should be detected and flagged.</li> <li>• Holidays affecting sales should be identified and marked</li> </ul> |                                                    |

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| <b>Story ID: US-03</b><br><b>Story Title: Feature Insights</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Priority: Medium</b><br><b>Estimated Hours: 6</b> |
| <b>User Story:</b><br><p>As a business manager, I want the system to generate meaningful features from sales data so that I can understand patterns and trends in business performance.</p> <b>Description:</b><br><p>This story involves generating analytical insights such as time-based patterns, trends, and rolling averages to help users understand business performance.</p> <b>Sub User Stories:</b> <ul style="list-style-type: none"> <li>• As a user, I want to view monthly and weekly sales patterns.</li> <li>• As a user, I want to analyze past sales trends.</li> <li>• As a user, I want to see rolling averages of sales.</li> </ul> <b>Acceptance Criteria:</b> <ul style="list-style-type: none"> <li>• The system should display monthly and weekly patterns.</li> <li>• Historical sales trends should be visualized.</li> <li>• Rolling averages should be calculated and shown</li> </ul> |                                                      |

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| <b>Story ID:</b> US-04<br><b>Story Title:</b> Data Visualization (EDA)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Priority:</b> High<br><b>Estimated Hours:</b> 5 |
| <b>User Story:</b><br><br>As a store owner, I want to see visual reports of sales trends and seasonality so that I can make informed business decisions.<br><br><b>Description:</b><br><br>This story focuses on providing visual exploratory data analysis (EDA) tools, including trend graphs, seasonal patterns, and downloadable reports.<br><br><b>Sub User Stories:</b> <ul style="list-style-type: none"><li>• As a user, I want to view trend graphs.</li><li>• As a user, I want to view seasonal patterns.</li><li>• As a user, I want to download visual reports.</li></ul><br><b>Acceptance Criteria:</b> <ul style="list-style-type: none"><li>• Trend graphs should be displayed clearly.</li><li>• Seasonal patterns should be visualized.</li><li>• Users should be able to download visual reports.</li></ul> |                                                    |

Sprint 2 focuses on building and evaluating prediction capabilities to provide accurate sales forecasts for better business decision-making.

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| <b>Story ID: US-05</b><br><b>Story Title: Access to Sales Predictions</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Priority: High</b><br><b>Estimated Hours: 5</b> |
| <b>User Story:</b><br><br>As a retail analyst<br>I want the system to generate sales predictions<br>So that I can estimate future sales trends and plan accordingly<br><br><b>Description:</b><br><br>This feature allows users to view predicted sales values based on historical data to support forecasting.<br><br><b>Sub User Stories:</b> <ul style="list-style-type: none"><li>• I want to view predicted sales values for upcoming dates</li><li>• I want predictions based on historical sales data</li><li>• I want predictions to update when new data is available .</li></ul><br><b>Acceptance Criteria:</b> <ul style="list-style-type: none"><li>• <b>Given</b> historical sales data is available<br/><b>When</b> the user requests predictions<br/><b>Then</b> the system should display future sales values</li><li>• <b>Given</b> new data is added<br/><b>When</b> predictions are generated again<br/><b>Then</b> updated predictions should be shown</li></ul> |                                                    |

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| <b>Story ID: US-06</b><br><br><b>Story Title: Model Accuracy Insights</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Priority: High</b><br><br><b>Estimated Hours: 7</b> |
| <b>User Story:</b><br><br>As a retail manager<br><b>I want</b> to see prediction accuracy<br><b>So that</b> I can trust the forecasts<br><br><b>Description:</b><br><br>This feature provides simple and clear insights into how accurate the predictions are.<br><br><b>Sub User Stories:</b> <ul style="list-style-type: none"> <li>• I want to view accuracy metrics in a simple format</li> <li>• I want to compare performance of different prediction approaches</li> <li>• I want visual summaries of prediction accuracy</li> </ul><br><b>Acceptance Criteria:</b> <ul style="list-style-type: none"> <li>• <b>Given</b> predictions are generated<br/> <b>When</b> the user views performance<br/> <b>Then</b> the system should display accuracy metrics</li> <li>• <b>Given</b> multiple prediction methods exist<br/> <b>When</b> the user checks performance<br/> <b>Then</b> a comparison view should be shown</li> </ul> |                                                        |



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| <b>Story ID: US-07</b><br><br><b>Story Title: Improved Prediction Quality</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Priority: Medium</b><br><br><b>Estimated Hours: 6</b> |
| <b>User Story:</b><br><br>As a business user<br>I want the system to provide highly accurate predictions<br>So that I can minimize risks in planning inventory and sales strategies<br><br><b>Description:</b><br><br>This story involves generating analytical insights such as time-based patterns, trends, and rolling averages to help users understand business performance.<br><br><b>Sub User Stories:</b> <ul style="list-style-type: none"> <li>• I want predictions to improve over time</li> <li>• I want the system to reduce forecasting errors</li> <li>• I want consistent and reliable prediction results</li> </ul><br><b>Acceptance Criteria:</b> <ul style="list-style-type: none"> <li>• <b>Given</b> predictions are generated<br/> <b>When</b> the system evaluates results<br/> <b>Then</b> prediction errors should be minimized</li> <li>• <b>Given</b> system improvements are applied<br/> <b>When</b> predictions are generated again<br/> <b>Then</b> accuracy should improve or remain consistent</li> </ul> |                                                          |

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| <b>Story ID: US-08</b><br><br><b>Story Title: Reliable Forecasting System</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Priority: High</b><br><br><b>Estimated Hours: 5</b> |
| <b>User Story:</b><br><br>As a retail decision-maker<br>I want the system to use the best available prediction method<br>So that I receive the most reliable sales forecasts<br><br><b>Description:</b><br><br>This ensures the system consistently uses the most effective method to provide reliable results.<br><br><b>Sub User Stories:</b> <ul style="list-style-type: none"> <li>• I want the system to automatically use the most accurate model</li> <li>• I want consistent prediction results across different scenarios</li> <li>• I want confidence that the system is using optimized methods</li> </ul> <b>Acceptance Criteria:</b> <ul style="list-style-type: none"> <li>• <b>Given</b> multiple prediction methods are available<br/> <b>When</b> predictions are generated<br/> <b>Then</b> the system should use the most accurate method</li> <li>• <b>Given</b> predictions are shown<br/> <b>When</b> the user views results<br/> <b>Then</b> outputs should be consistent and reliable</li> </ul> |                                                        |

Sprint 3 focused on the complete development and integration of the Retail Intelligence platform by implementing both frontend and backend functionalities. This sprint delivered a fully functional web-based system with interactive dashboards, user management, analytics, recommendations, and support features for end users.

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| <b>Story ID: US-09</b><br><br><b>Story Title: Secure User Access</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Priority: High</b><br><br><b>Estimated Hours: 2</b> |
| <b>User Story:</b><br><br>As a retail system user<br>I want to securely register and log into the platform<br>So that I can safely access forecasting tools and business insights<br><br><b>Description:</b><br>This ensures authorized users can securely access the system and use personalized dashboard features while protecting business data.<br><br><b>Sub User Stories:</b> <ul style="list-style-type: none"><li>• I want to create my account to use the platform</li><li>• I want to log in securely using my credentials</li><li>• I want only authorized users to access system data</li></ul><br><b>Acceptance Criteria:</b> <ul style="list-style-type: none"><li>• Given a new user wants to access the platform<br/>When registration details are submitted<br/>Then the system should create a user account successfully</li><li>• Given a registered user enters valid credentials<br/>When logging into the system<br/>Then the user should be granted secure access to the dashboard</li><li>• Given invalid credentials are entered<br/>When login is attempted<br/>Then the system should deny access and display an error message</li></ul> |                                                        |

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| <b>Story ID: US-10</b><br><br><b>Story Title: Interactive Dashboard Overview</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Priority: High</b><br><br><b>Estimated Hours: 8</b> |
| <b>User Story:</b><br>As a retail decision-maker<br>I want to view summarized retail insights in one dashboard<br>So that I can quickly understand business performance<br><br><b>Description:</b><br>This provides users with a centralized dashboard displaying sales, products, promotions, and forecasting insights for better decision-making.<br><br><b>Sub User Stories:</b> <ul style="list-style-type: none"> <li>• I want to see total sales and product statistics</li> <li>• I want to view forecast summaries visually</li> <li>• I want quick access to important business metrics</li> </ul><br><b>Acceptance Criteria:</b> <ul style="list-style-type: none"> <li>• Given the user opens the dashboard<br/> When system data loads<br/> Then summary cards and charts should display key retail metrics</li> <li>• Given updated sales data exists<br/> When the dashboard refreshes<br/> Then the displayed metrics should reflect the latest data</li> </ul> |                                                        |

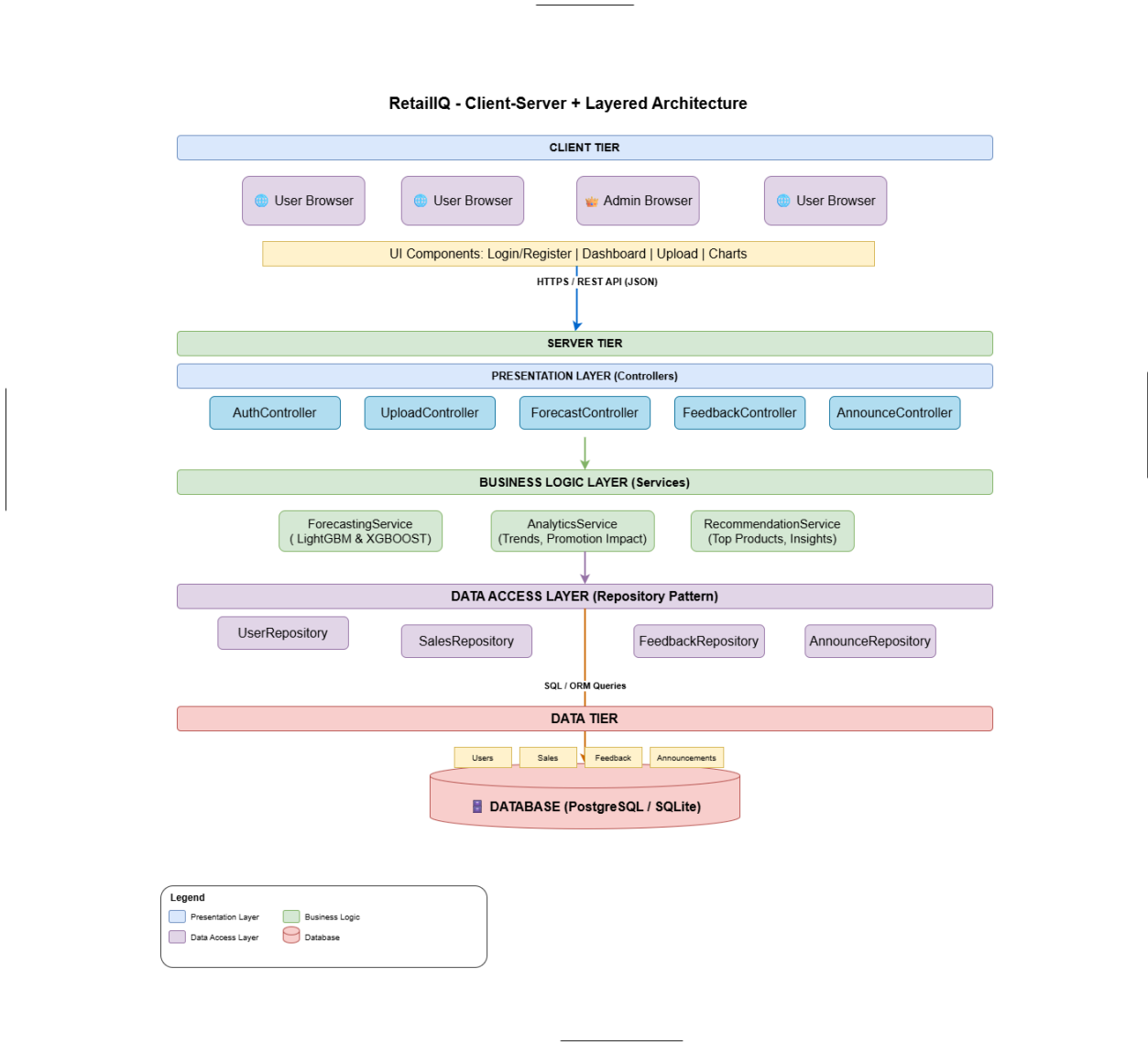
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| <b>Story ID: US-11</b><br><br><b>Story Title: Sales Performance Analysis</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Priority: High</b><br><br><b>Estimated Hours: 5</b> |
| <b>User Story:</b><br>As a retail analyst<br>I want to visualize historical sales trends and promotional impact<br>So that I can identify demand patterns and sales opportunities<br><br><b>Description:</b><br>This allows users to analyze sales performance through charts, trend filters, and promotion-based comparisons.<br><br><b>Sub User Stories:</b> <ul style="list-style-type: none"> <li>• I want to view sales trends using charts</li> <li>• I want to compare promotional and non-promotional sales</li> <li>• I want to filter data by different time periods</li> </ul> <b>Acceptance Criteria:</b> <ul style="list-style-type: none"> <li>• Given sales data is available<br/> When the user opens analytics pages<br/> Then trend charts should display historical sales performance</li> <li>• Given time filter options are selected<br/> When the user changes the period<br/> Then the system should update charts accordingly</li> <li>• Given promotion data exists<br/> When promotional analysis is viewed<br/> Then the system should show sales differences during promotions</li> </ul> |                                                        |

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| <b>Story ID: US-12</b><br><br><b>Story Title: Product Insights and Recommendations</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Priority: High</b><br><br><b>Estimated Hours: 7</b> |
| <b>User Story:</b><br>As a retail manager<br>I want to identify top products and receive recommendations<br>So that I can improve inventory and sales strategies<br><br><b>Description:</b><br>This helps users recognize high-performing products and receive system-generated recommendations for better business decisions.<br><br><b>Sub User Stories:</b> <ul style="list-style-type: none"> <li>• I want to see top-selling products</li> <li>• I want to understand product performance rankings</li> <li>• I want recommendations for inventory planning</li> </ul> <b>Acceptance Criteria:</b> <ul style="list-style-type: none"> <li>• Given product sales data is available<br/> When the user opens the product page<br/> Then the top-performing products should be displayed</li> <li>• Given system analysis is completed<br/> When recommendations are generated<br/> Then useful business suggestions should be shown</li> </ul> |                                                        |

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| <b>Story ID: US-13</b><br><br><b>Story Title: Communication and Support System</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Priority: High</b><br><br><b>Estimated Hours: 6</b> |
| <b>User Story:</b><br>As a platform user<br>I want access to updates, support, and feedback options<br>So that I can stay informed and communicate with administrators<br><br><b>Description:</b><br>This enables users to receive announcements, read industry insights, submit feedback, and access help resources.<br><br><b>Sub User Stories:</b> <ul style="list-style-type: none"> <li>• I want to read important system announcements</li> <li>• I want to access retail news and insights</li> <li>• I want to provide feedback to improve the system</li> <li>• I want to use the help guide when needed</li> </ul> <b>Acceptance Criteria:</b> <ul style="list-style-type: none"> <li>• Given administrators post updates<br/>When users open announcements<br/>Then the latest system information should be displayed</li> <li>• Given the user submits feedback<br/>When the form is completed<br/>Then feedback should be stored successfully</li> <li>• Given a new user needs assistance<br/>When the help page is accessed<br/>Then usage instructions should be available</li> </ul> |                                                        |

# Design of System

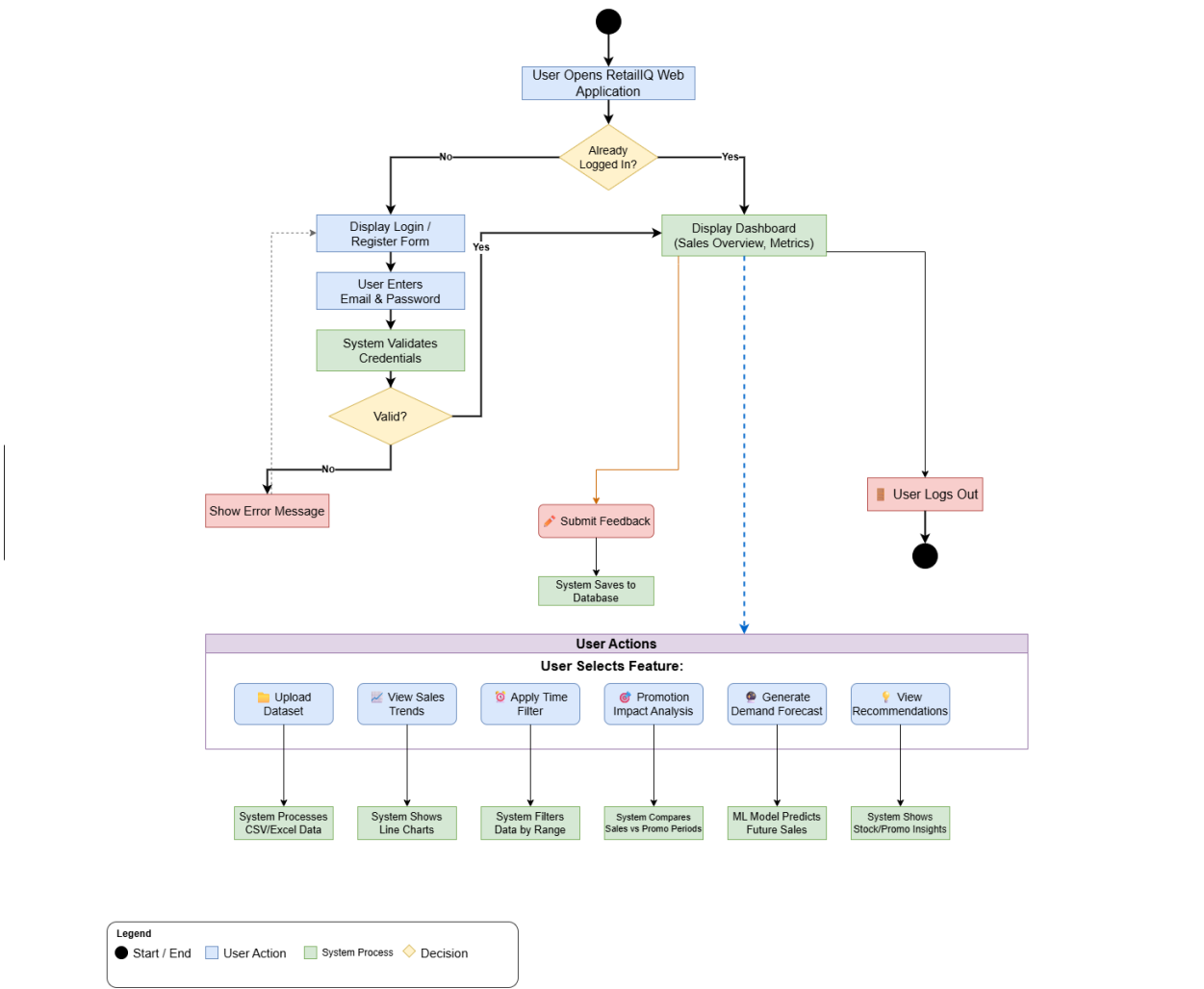
Architecture Diagram:



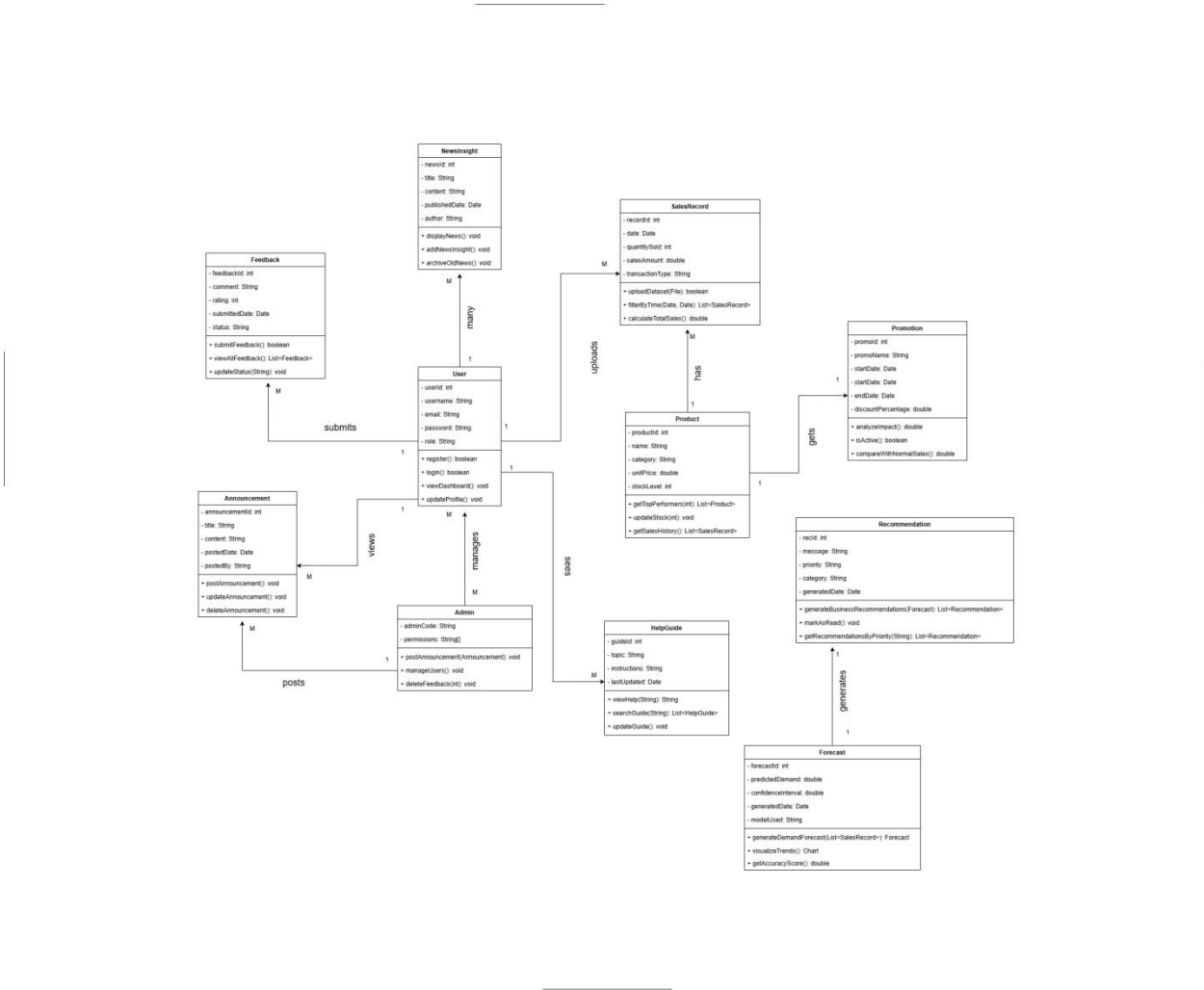


Activity Diagram

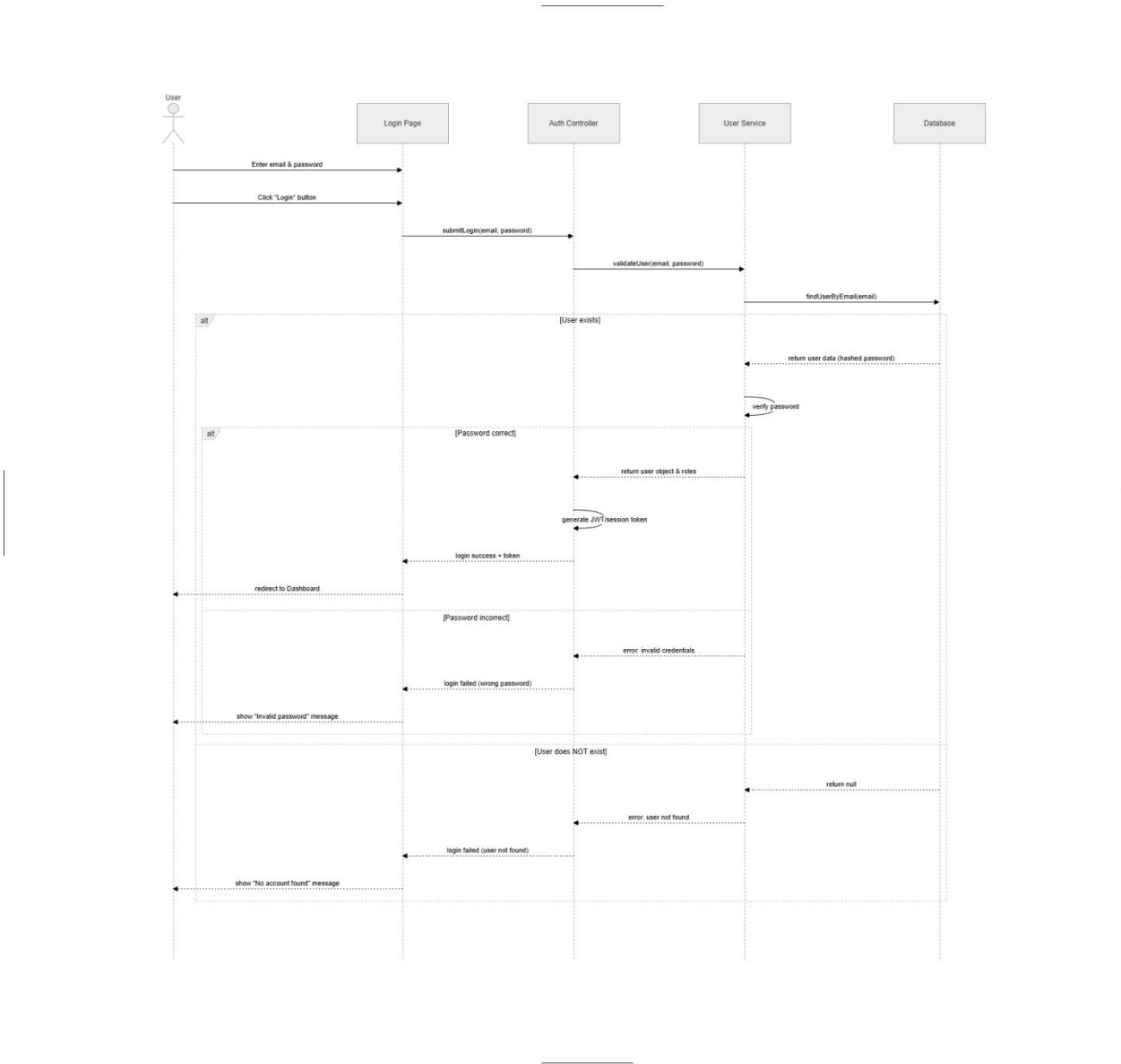
RetailIQ - Activity Diagram (User Workflow)



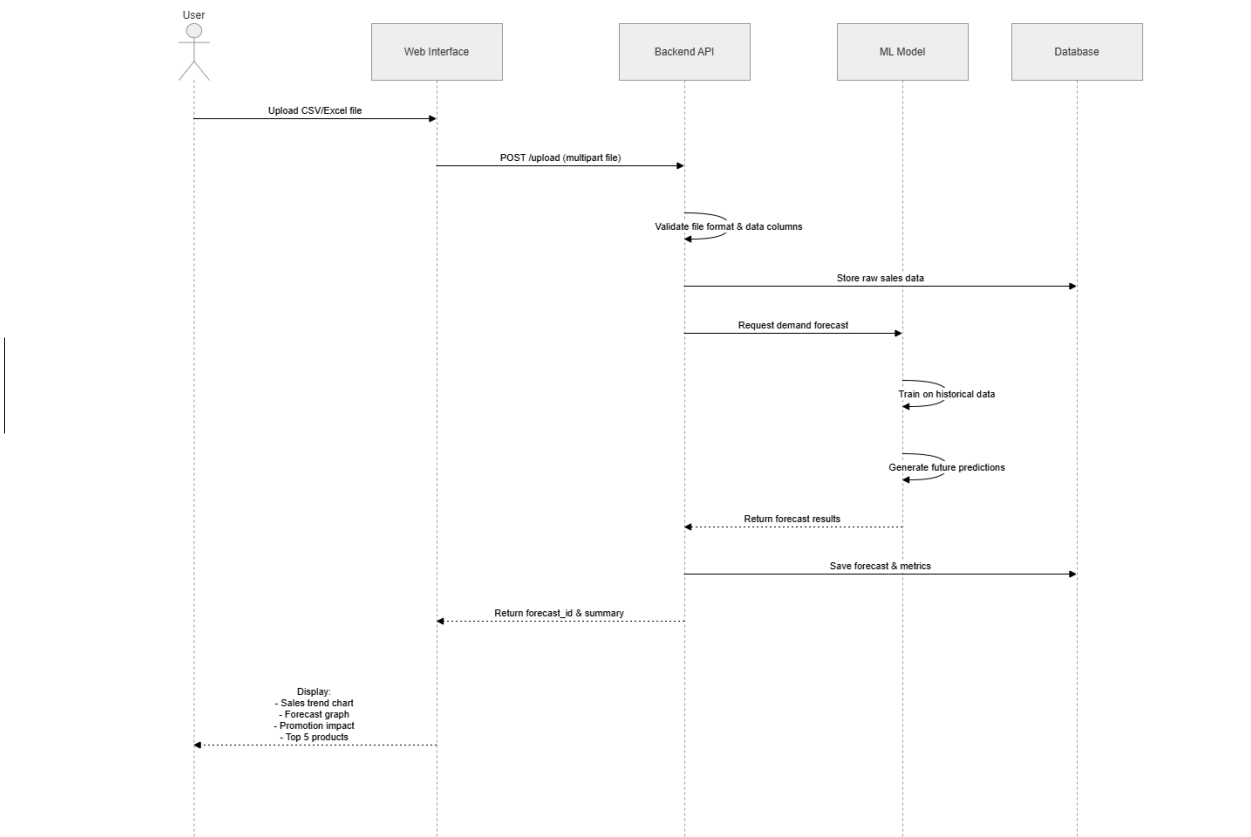
Class Diagram



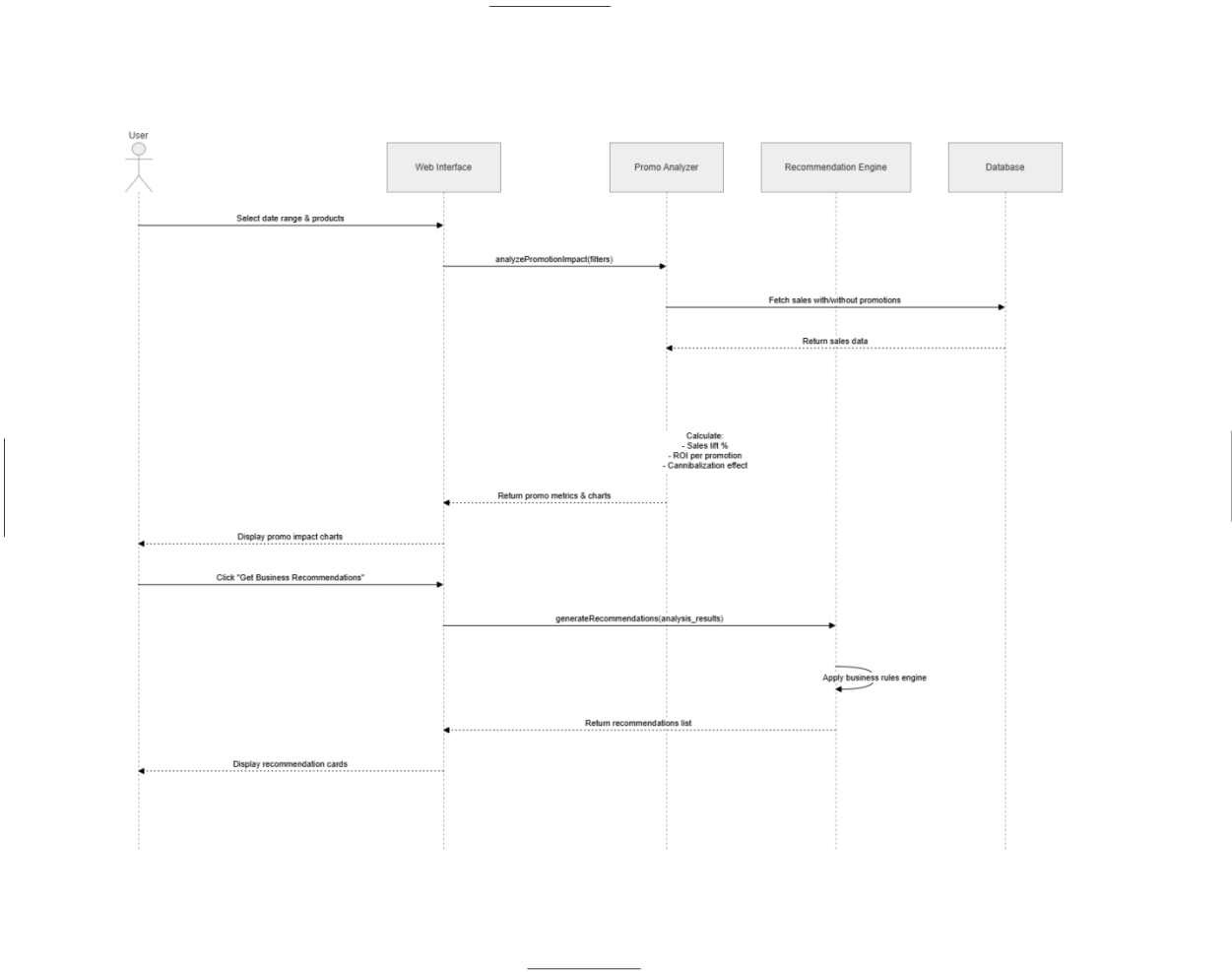
# Sequence Diagram from Login



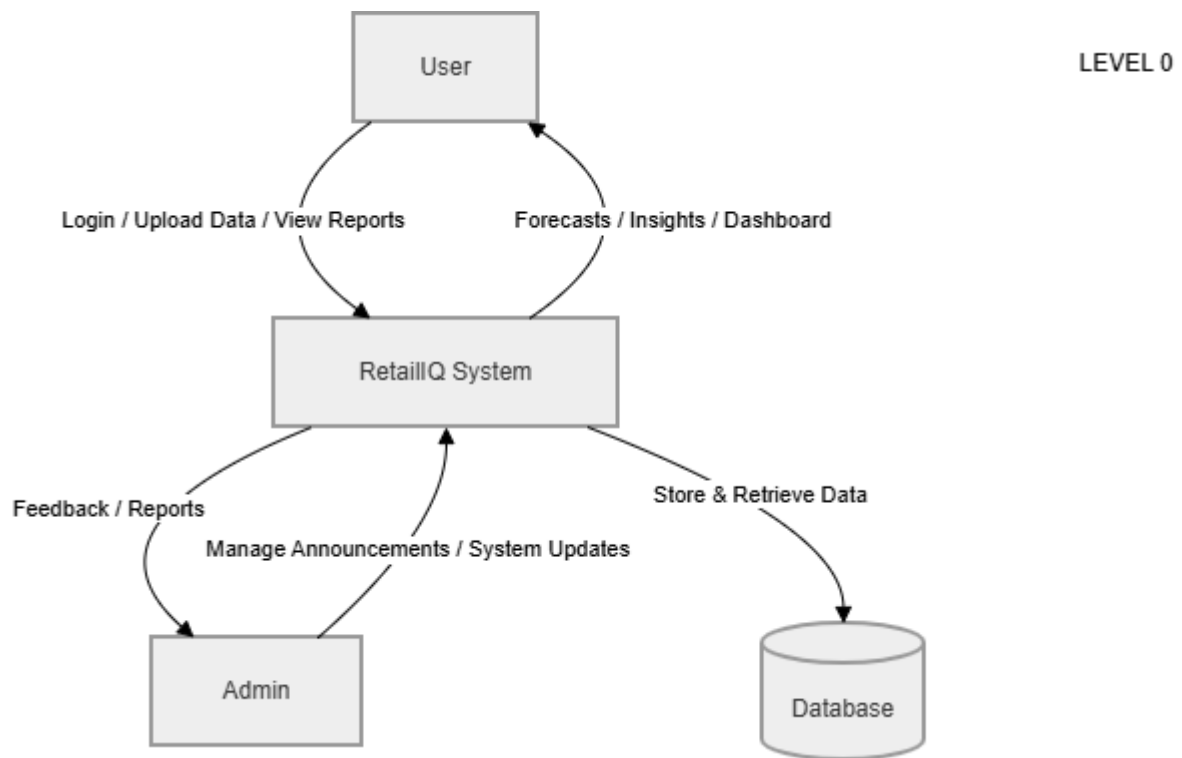
# Sequence Diagram for Data Set Upload and Forecast



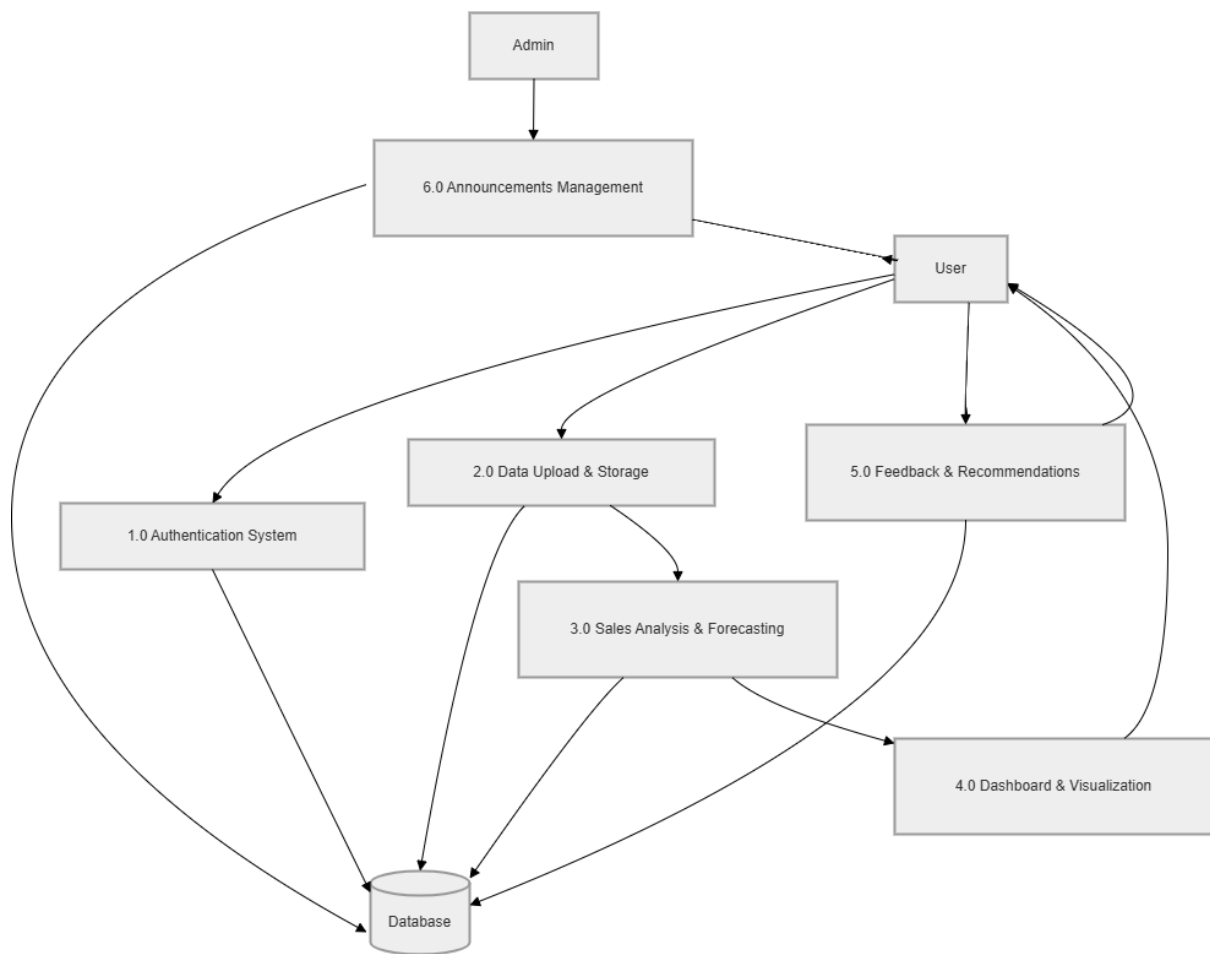
# Sequence Diagram for Sales Prediction



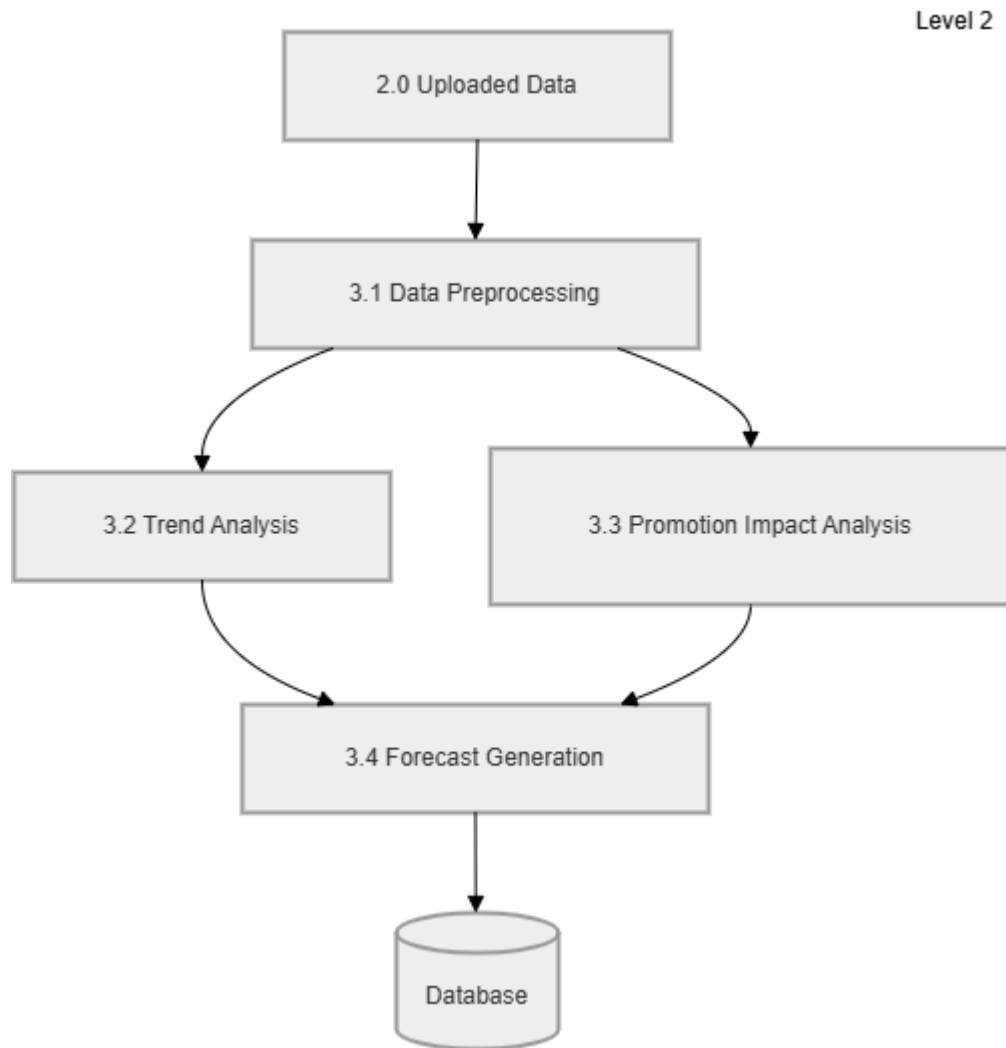
## DFD Level 0



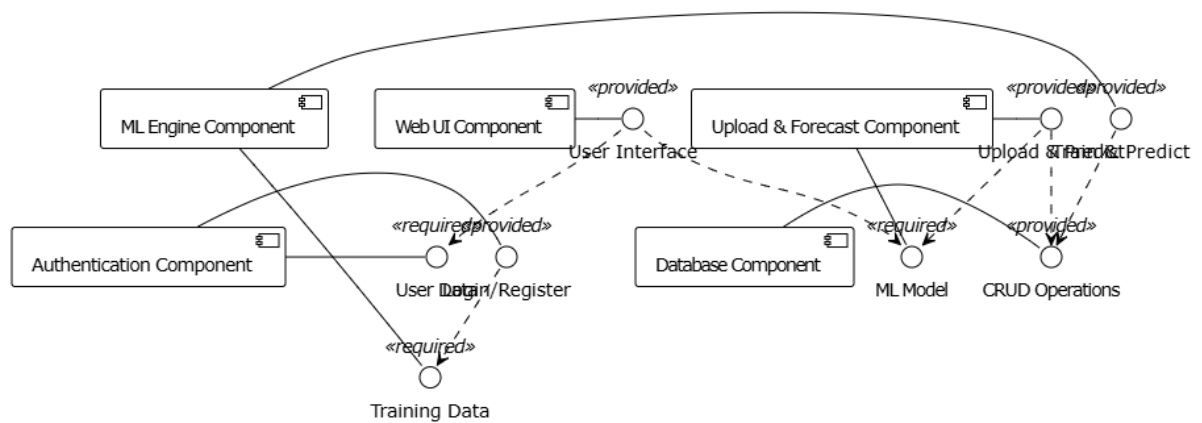
## DFD Level 1



## DFD Level 2

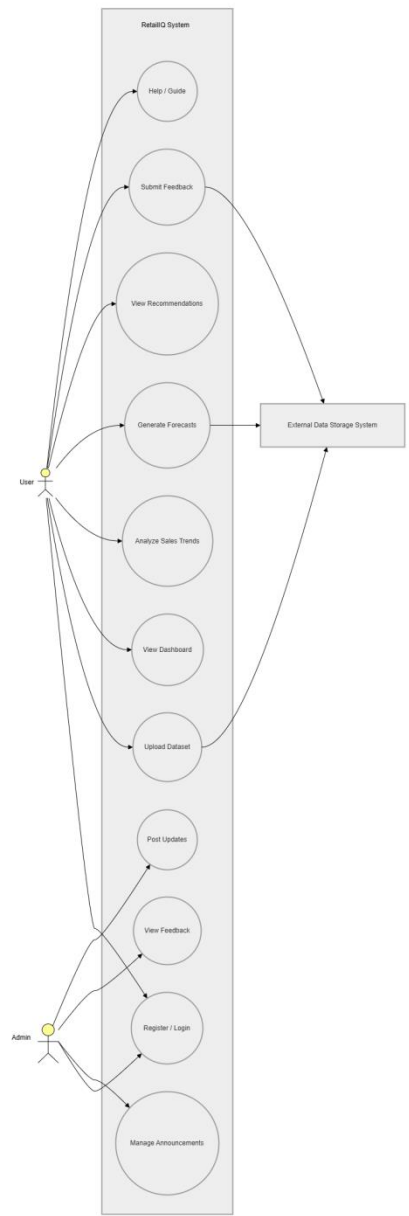


## Component Diagram

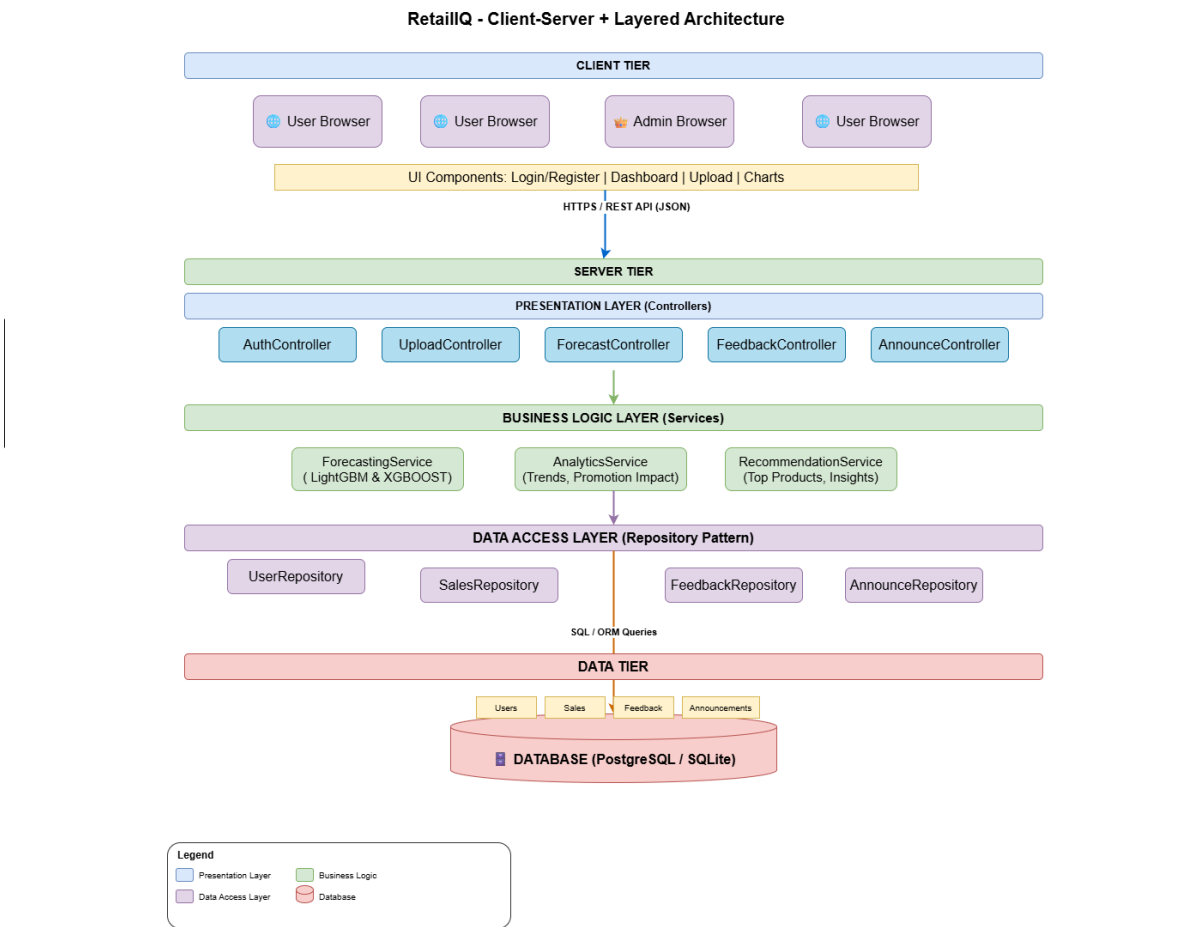




Use Case



# Architecture Style



## Architecture Pattern

The RetailIQ system follows a **Client-Server Architecture combined with Layered Architecture** to ensure modularity, maintainability, and scalability. This architectural approach enables efficient communication between users and the system while organizing the internal functionalities into structured layers.

The **Client-Server Architecture** manages communication between the frontend interface (client) and backend services (server), while the **Layered Architecture** organizes server-side components into logical layers such as presentation, business logic, and data management.

This combination was selected because RetailIQ is a web-based retail intelligence platform that requires secure user interaction, efficient data processing, forecasting capabilities, and structured information management.

# Client-Server Architecture

RetailIQ operates using the client-server model, where users interact with the system through a web interface, and all business processing is handled by centralized server services.

## Client Side

The client represents the user-facing graphical interface developed for system interaction.

Responsibilities include:

- User registration and login
- Dashboard visualization
- Viewing sales trends and promotion analysis
- Accessing product recommendations
- Submitting feedback
- Viewing announcements and news
- Using help and support features

## Technology Used:

- React.js
- Bootstrap
- Chart Libraries

## Server Side

The server processes requests received from clients and provides responses after executing business logic.

Responsibilities include:

- User authentication and authorization
- Sales data processing
- Recommendation generation
- Dashboard metric calculations
- Feedback handling
- Announcement management
- Forecasting and analytics services
- Database communication

## Technology Used:

- Backend APIs
- Business logic modules
- Machine learning forecasting models

# Layered Architecture

Within the server, RetailIQ follows layered architecture to separate system responsibilities into independent layers.

## Presentation Layer

This layer provides the graphical user interface for end users.

Components include:

- Login and signup pages
- Dashboard
- Product pages
- Analytics pages
- Feedback page
- Announcements page
- Help and guide pages

## Business Logic Layer

This layer processes user requests and implements the application's core functionalities.

Components include:

- Authentication management
- Dashboard calculations
- Sales trend analysis
- Promotion impact analysis
- Product ranking
- Recommendation engine
- Forecasting services
- Feedback management

## Data Layer

This layer stores, retrieves, and manages application data.

Components include:

- User data
- Retail sales records
- Product information
- Feedback records
- Announcement data
- Forecasting model files

### Technology Used:

- Database management system
- Serialized machine learning models

### Why This Architecture Was Selected

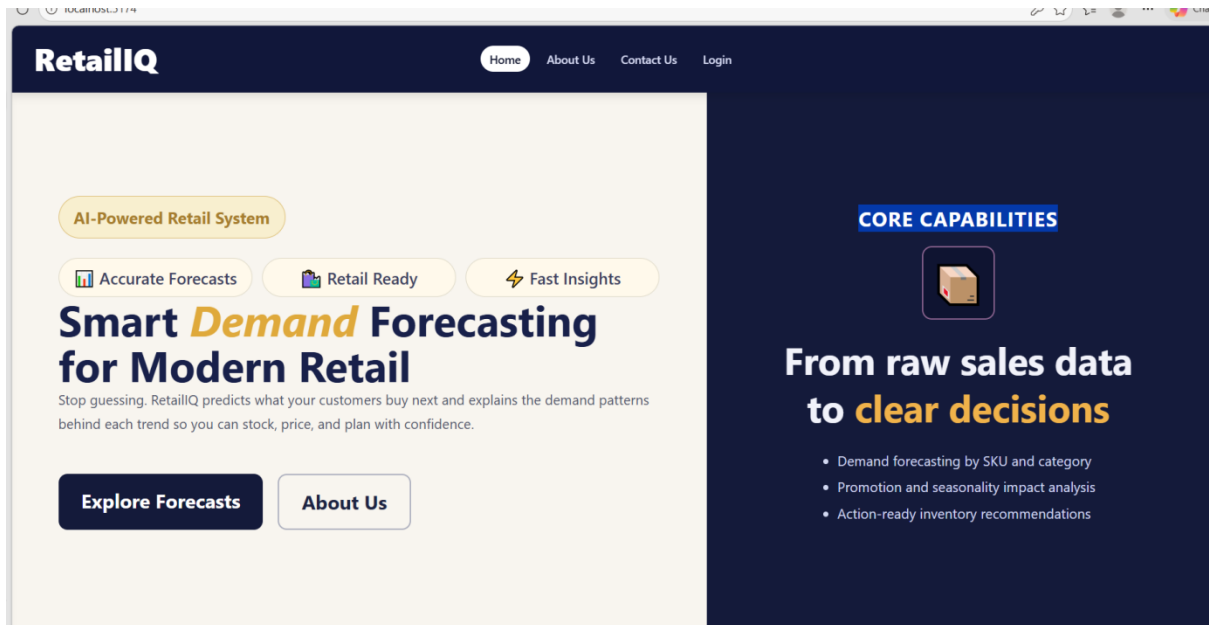
The combined client-server and layered architecture was selected because it offers several benefits for RetailIQ:

- **Separation of concerns** by dividing functionalities into layers
- **Scalability** for future addition of new features
- **Maintainability** by allowing independent updates to system modules
- **Security** through centralized authentication and data access
- **Reusability** of backend services across multiple interfaces
- **Improved performance** through efficient client-server communication

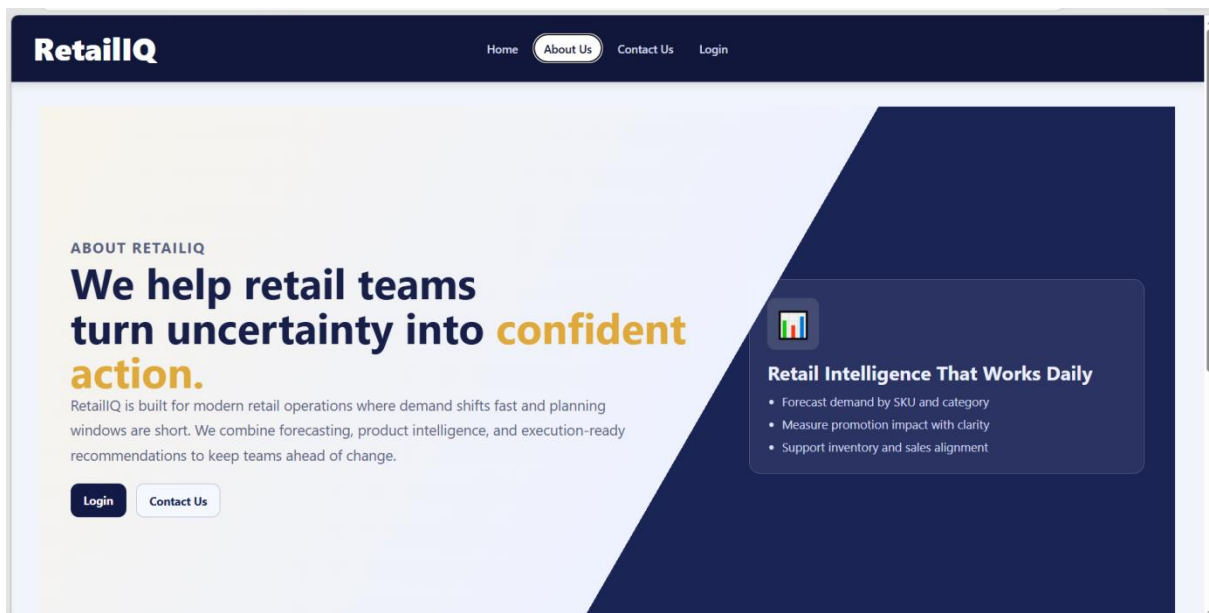
This architecture supports RetailIQ's objective of delivering a reliable and interactive retail intelligence platform.

## Actual implementation screenshots

Main Homepage:



About Us:



Sign Up:

RetailIQ

[Home](#)[About Us](#)[Contact Us](#)[Login](#)

Create Your RetailIQ Account.

Join the platform and start forecasting with precision.

Account Registration

Full Name \*

Enter your full name

Phone Number \*

03XXXXXXXX

Email Address \*

example@gmail.com

Password \*

Create password

Confirm Password \*

Re-enter password

Create Account

Platform Benefits: Access real-time insights, customized alerts, team collaboration.

Login:

RetailIQ

[Home](#)[About Us](#)[Contact Us](#)[Login](#)

RetailIQ



RETAILIQ - Intelligent Forecasting.

Welcome Back to RetailIQ.

Sign in to access your forecasts.

Email Address

rubab@gmail.com

Password

\*\*\*\*\*

Sign In

[Forgot Password?](#)

Don't have an account? [Register](#)

RetailIQ

[Home](#)[About Us](#)[Contact Us](#)[Login](#)

Reset

Password

Enter your new password below.

New Password

\*\*\*\*\*

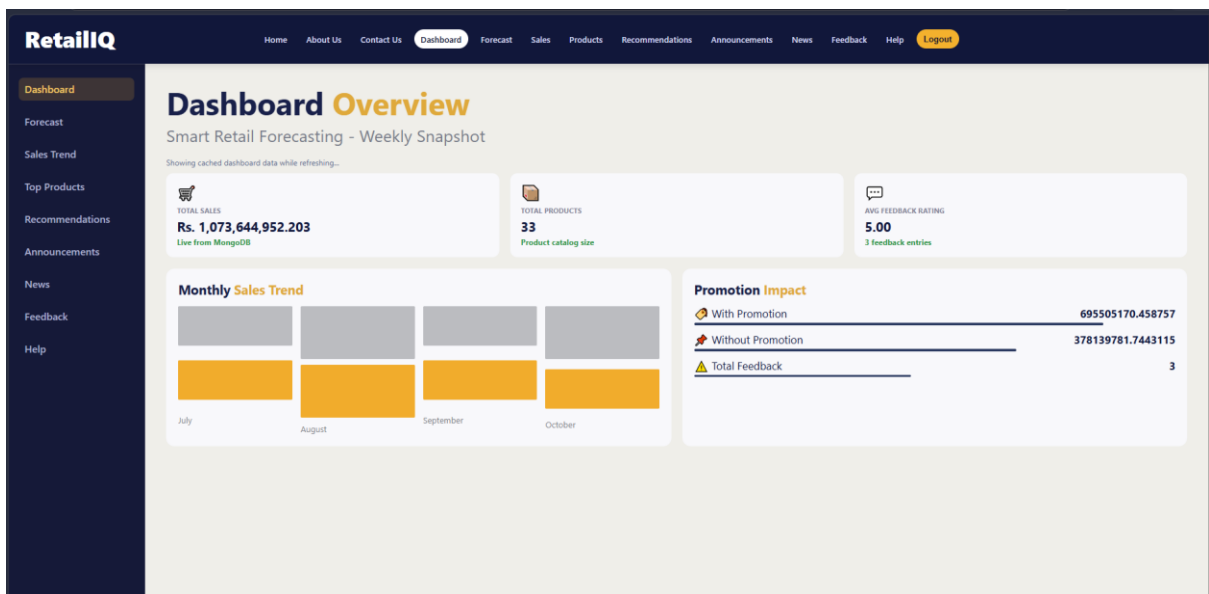
Confirm Password

\*\*\*\*\*

Reset Password

Back to Login

Dashboard:





Dashboard

Forecast

Sales Trend

Top Products

Recommendations

Announcements

News

Feedback

Help

Home

About Us

Contact Us

Dashboard

Forecast

Sales

Products

Recommendations

Announcements

News

Feedback

Help

Logout

## Sales Forecast

Enter store and product family details to get inventory guidance from prediction bands.

### Forecast Inputs

Store Number

8

Family

BREAD/BAKERY

On Promotion

No (0)

Date

01/15/2017

Predict Sales

### Moderate Sales

Moderate sales expected. Keep balanced inventory for steady movement.

Range logic: 0 or less = very low, 0-100 = low, 100-1000 = moderate, 1000-2000 = good, 2000+ = high.

796

RetailIQ

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[Forecast](#)
[Sales](#)
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Dashboard

Forecast

Sales Trend

Top Products

Recommendations

Announcements

News

Feedback

Help

Top Products & Performance

Data-driven rankings of your best-selling items.

Showing cached products while refreshing...

TOP PERFORMER

13

TOP SALES VALUE

343,462,734.886

TOTAL RANKED PRODUCTS

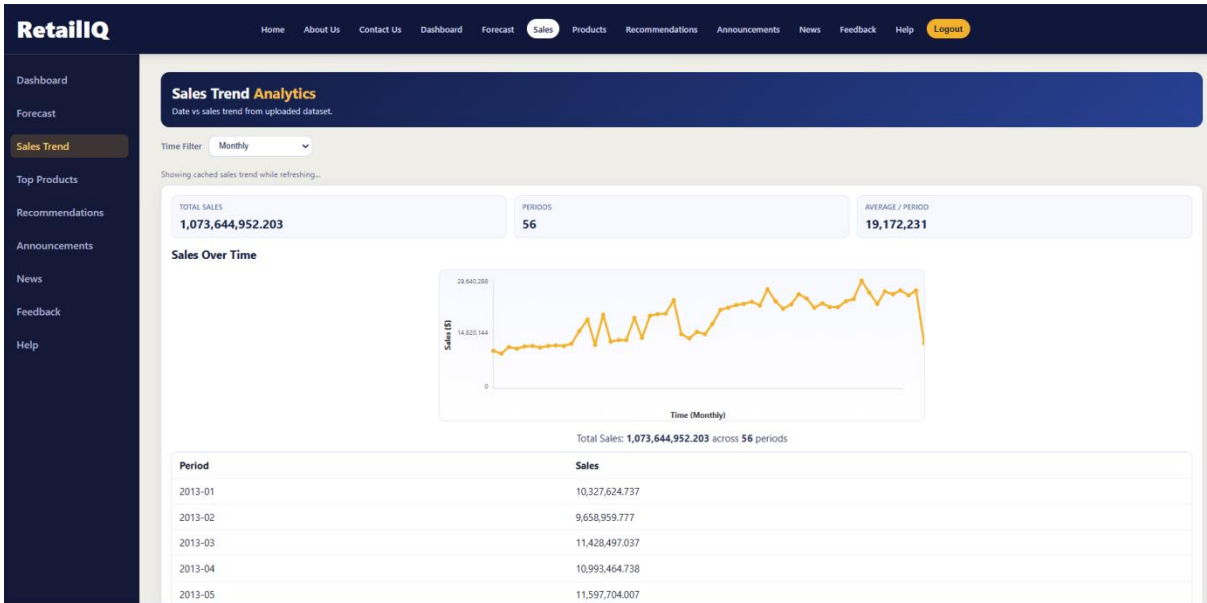
5

| PRODUCT NAME | TOTAL SALES     | RANK |
|--------------|-----------------|------|
| 13           | 343,462,734.886 | #1   |
| 4            | 216,954,486     | #2   |
| 31           | 122,704,684.676 | #3   |
| 8            | 97,521,289      | #4   |
| 9            | 64,487,709      | #5   |

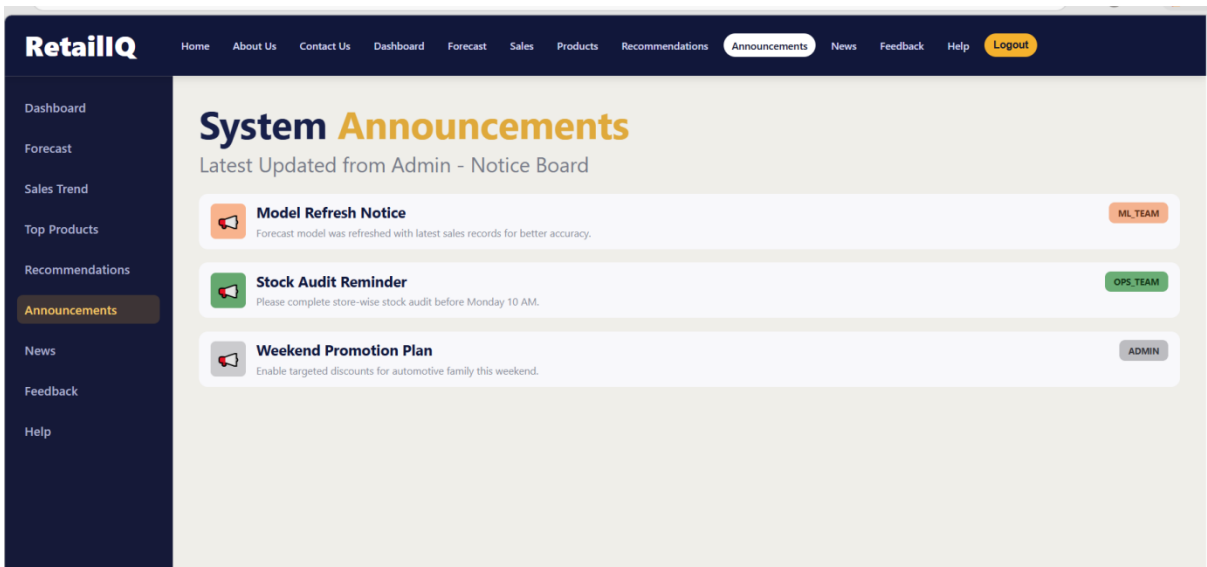
Top 5 Sales Comparison

|    |                 |
|----|-----------------|
| 13 | 343,462,734.886 |
| 4  | 216,954,486     |
| 31 | 122,704,684.676 |
| 8  | 97,521,289      |
| 9  | 64,487,709      |

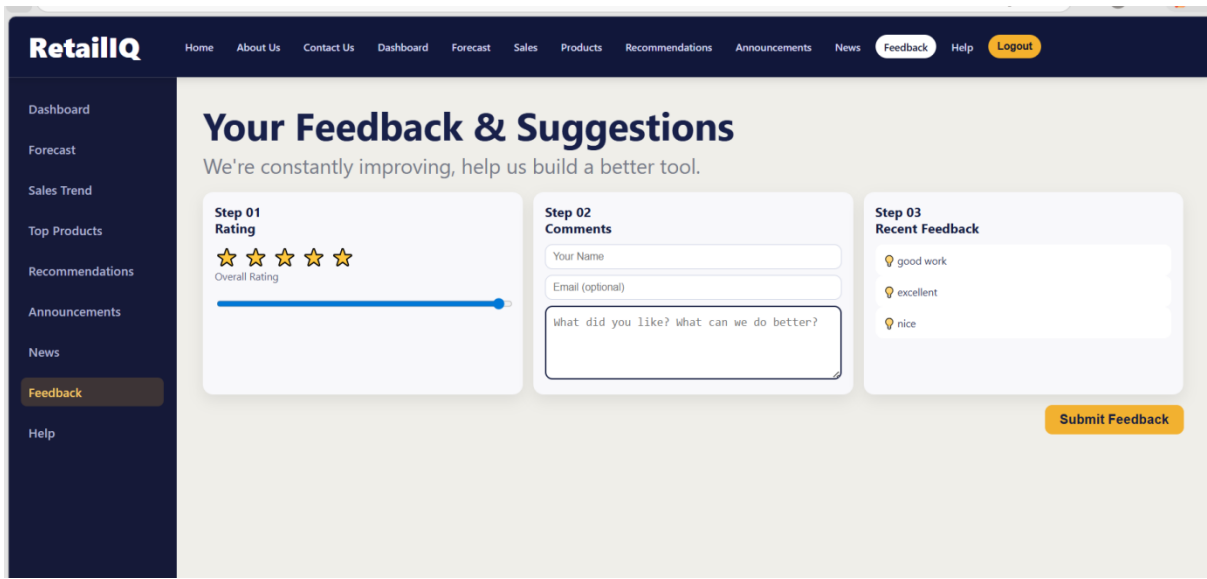
Sales Trend:



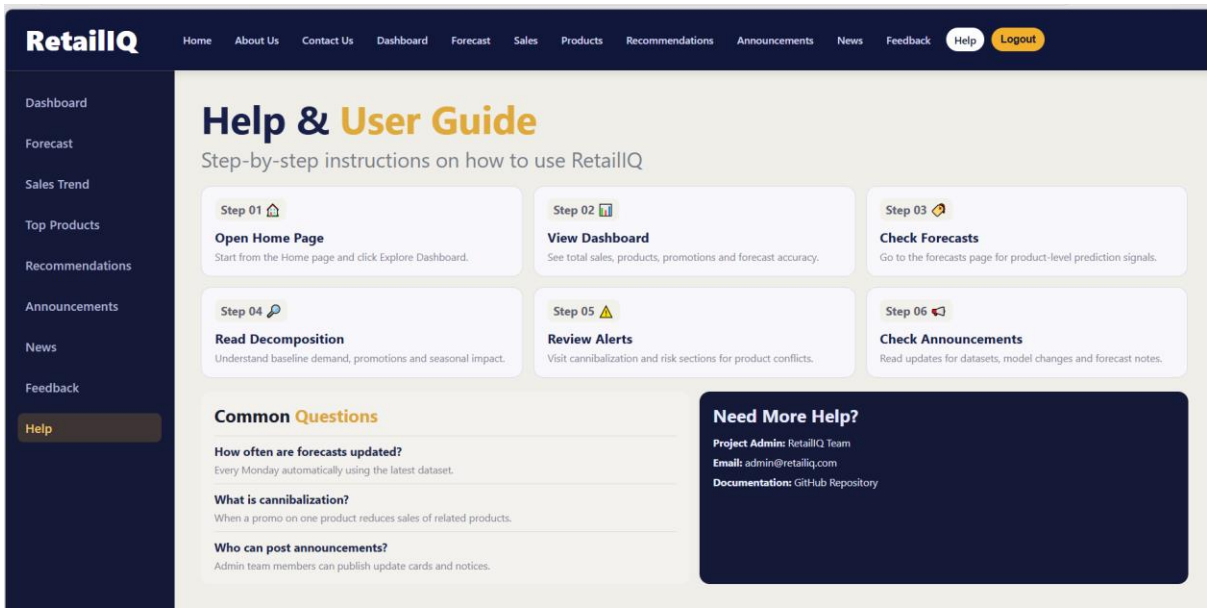
Announcements:



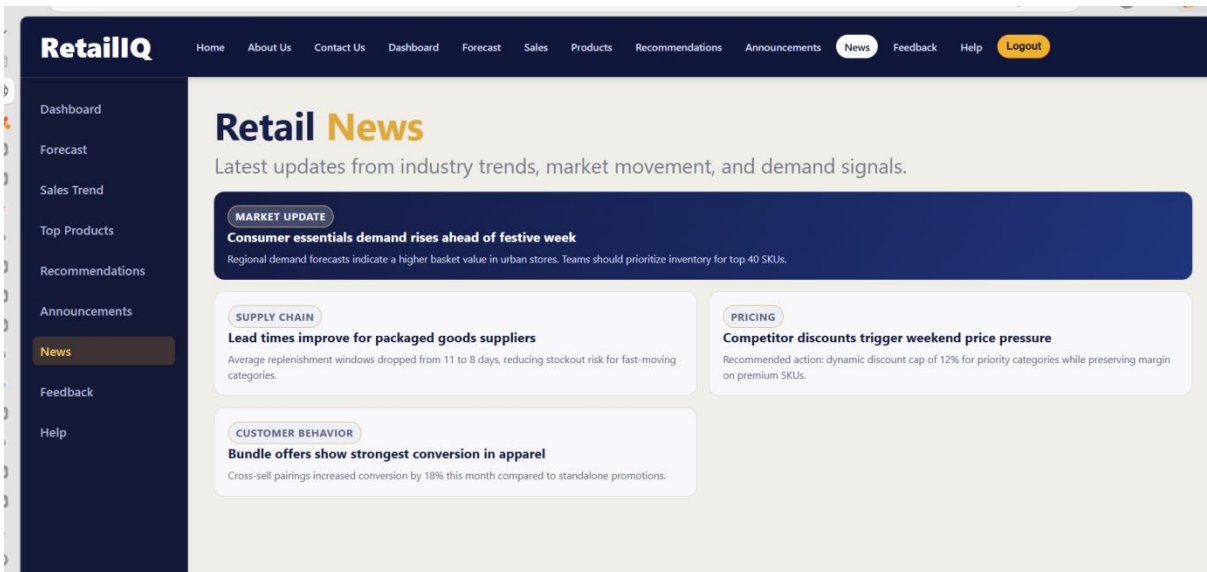
Feedback:



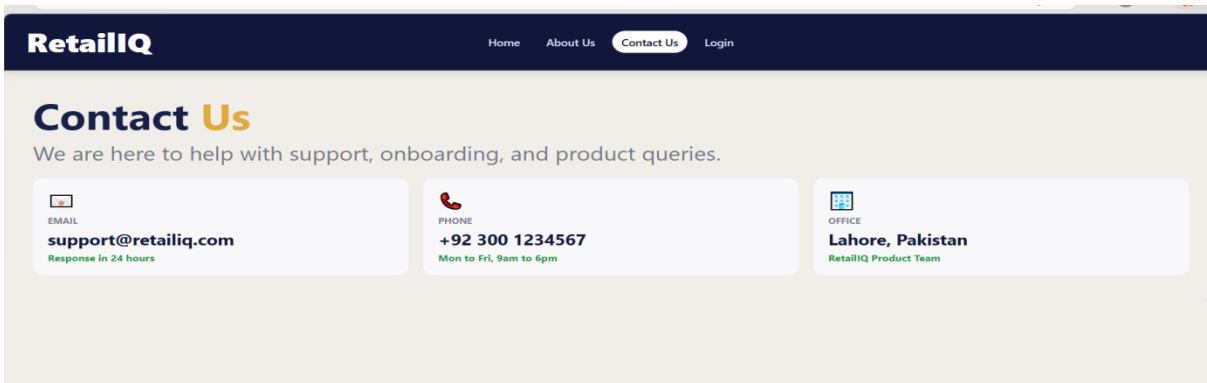
Help/User Guide:



News/Insights:

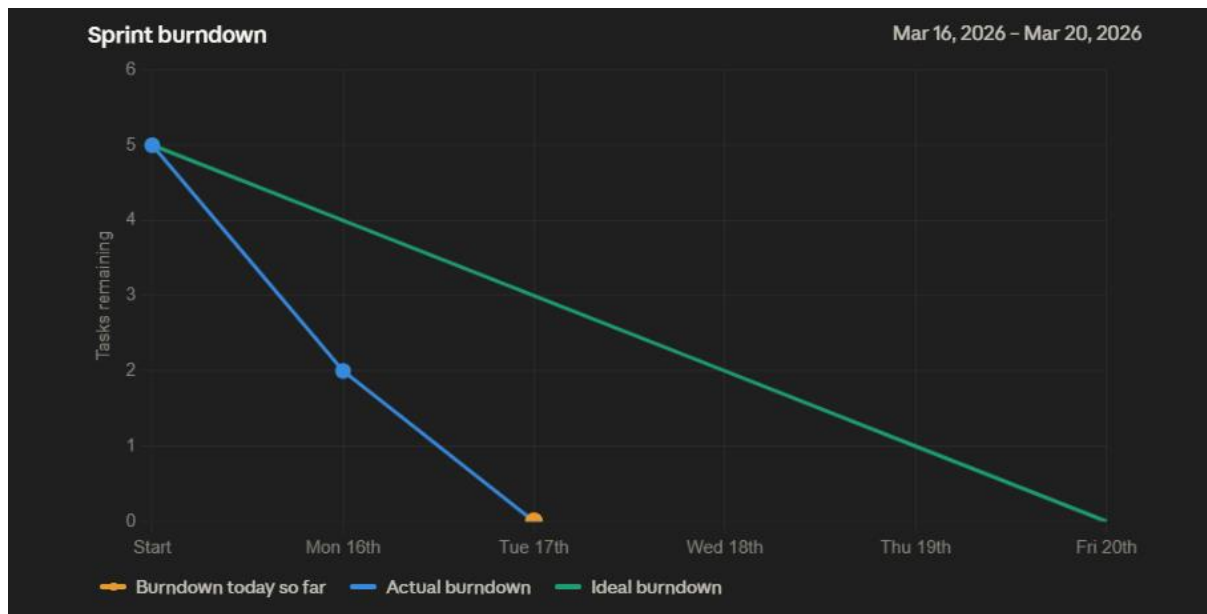


Contact us:

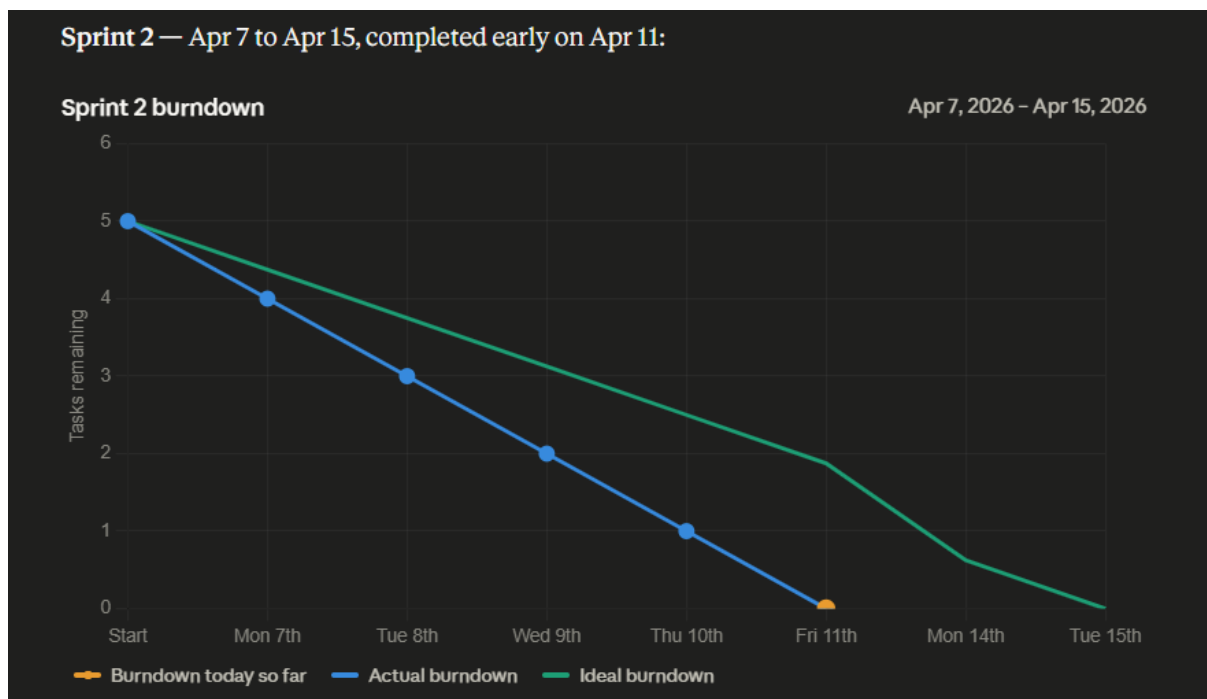


## Product Burn down chart for the project

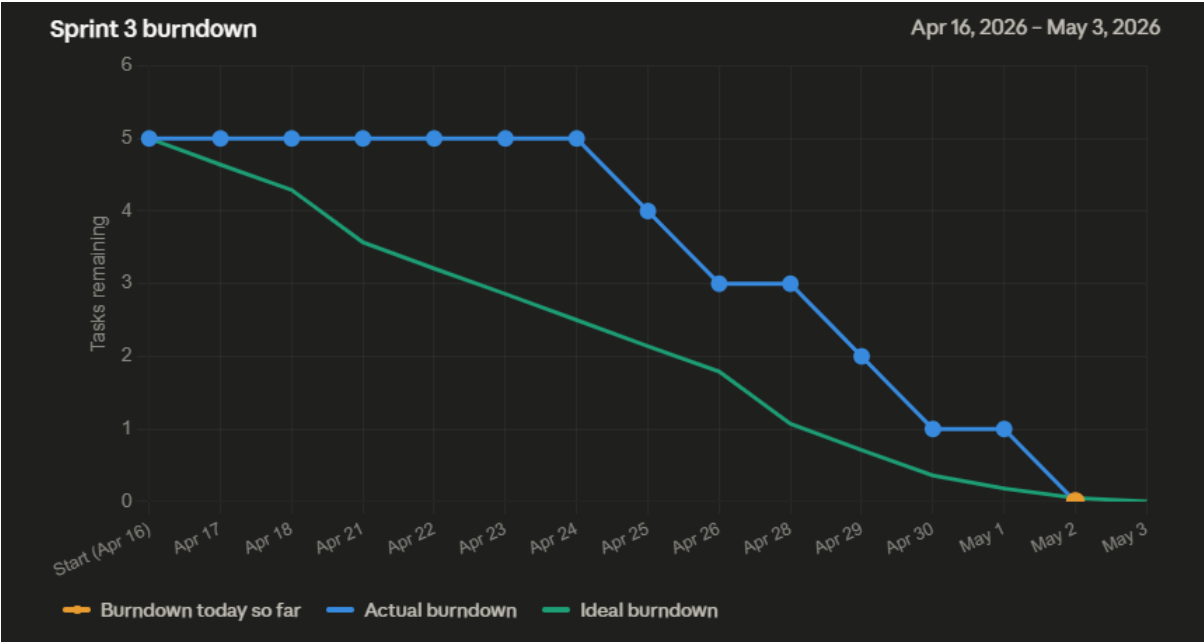
Sprint 1:



Sprint 2:



Sprint 3:



# Trello board screen shots

## Sprint 3 Trello Board

Spaces

RetailIQ

...

Summary

Timeline

Backlog

Board

Calendar

List

Forms

Goals

Development

Code

Archived work items

Docs

Shortcuts

Search board

AL

Filter

Complete sprint

TO DO 6

Sprint 3 - Web Work

RET-7

10

developing backend

RET-14

10

testing backend apis

RET-15

10

developing frontend

RET-16

10

integration of frontend and backend

RET-17

10

testing

RET-18

10

+ Create

IN PROGRESS

DONE

+

Spaces

RetailIQ

...

Summary

Timeline

Backlog

Board

Calendar

List

Forms

Goals

Development

Code

More

3

+

Search board

AL

Filter

Complete sprint

Group

TO DO 4

Sprint 3 - Web Work

RET-7

10

developing frontend

RET-16

10

integration of frontend and backend

RET-17

10

testing

RET-18

10

+ Create

IN PROGRESS 2

developing backend

RET-14

10

testing backend apis

RET-15

10

DONE

+

QU

Spaces

RetailIQ  ...



 Summary  Timeline  Backlog  Board  Calendar  List  Forms  Goals  Development  Code More  3 +

 Search board



 Filter

Complete sprint



Group 



TO DO 2

Sprint 3 - Web Work

☒ RET-7 10 

testing

☒ RET-18 10 

+ Create

IN PROGRESS 2

developing frontend

☒ RET-16 10 

integration of frontend and backend

☒ RET-17 10 

DONE 2 

developing backend

☒ RET-14  10 

testing backend apis

☒ RET-15  10 

+

 Quickstart

Spaces

RetailIQ  ...



 Summary  Timeline  Backlog  Board  Calendar  List  Forms  Goals  Development  Code More  3 +

 Search board



 Filter

Complete sprint



Group 



TO DO 1

Sprint 3 - Web Work

☒ RET-7 10 

+ Create

IN PROGRESS 1

testing

☒ RET-18 10 

DONE 4 

developing backend

☒ RET-14  10 

testing backend apis

☒ RET-15  10 

developing frontend ...

☒ RET-16  10 

integration of frontend and backend

☒ RET-17  10 

+

 Quickstart



Spaces

RetailIQ

...

Summary

Timeline

Backlog

Board

Calendar

List

Forms

Goals

Development

Code

Archived work items

Dr

Search board

AL

Filter

TO DO

+ Create

IN PROGRESS

DONE 6

Sprint 3 - Web Work

RET-7

✓

10

developing backend

RET-14

✓

10

testing backend apis

RET-15

✓

10

developing frontend

RET-16

✓

10

integration of frontend and backend

RET-17

✓

10

testing

RET-18

✓

10

Spaces

RetailIQ

...

Summary

Timeline

Backlog

Board

Calendar

List

Forms

Goals

Development

Code

More 3

+

Search calendar

Assignee

Type

Status

More filters

Today

< May 2026 >

Month

| Mon               | Tue | Wed | Thu | Fri   |
|-------------------|-----|-----|-----|-------|
| 27                | 28  | 29  | 30  | May 1 |
| RetailIQ Sprint 3 |     |     |     |       |
| 5                 | 6   | 7   | 8   |       |
| 11                | 12  | 13  | 14  | 15    |

SPRINT

...

CLOSED

RetailIQ Sprint 3

Start date

Complete date

Apr 24, 2026

May 03, 2026

# Boundary value analysis testing

## Introduction

Boundary Value Analysis (BVA) is a black-box software testing technique used to validate input fields by testing values at their minimum, maximum, and near-boundary limits. Since input-related errors commonly occur at boundary conditions, BVA helps ensure that the system correctly accepts valid inputs and rejects invalid ones.

For RetailIQ, boundary value analysis was performed on the **Sign Up** and **Login** interfaces to verify input validation, data constraints, and system reliability. The testing focused on fields such as **username, email, and password**, ensuring that the system behaves correctly for both valid and invalid boundary values.

### Sign Up Interface Testing

The Sign Up module allows users to register by providing required details. Boundary testing was conducted on all input fields.

### Fields Selected for Testing

| Field        | Validation Rule                |
|--------------|--------------------------------|
| Username     | 3 to 20 characters             |
| Email        | Must follow valid email format |
| Password     | 8 to 16 characters             |
| Phone Number | Exactly 11 digits              |

### Username Field Boundary Testing

| Test Case ID | Input                           | Boundary Type | Expected Result         | Actual Result   | Status |
|--------------|---------------------------------|---------------|-------------------------|-----------------|--------|
| SU-01        | ab                              | Below Minimum | Error message displayed | Error displayed | Pass   |
| SU-02        | abc                             | Minimum       | Accepted                | Accepted        | Pass   |
| SU-03        | abcdefghijklhijklmno (20 chars) | Maximum       | Accepted                | Accepted        | Pass   |
| SU-04        | abcdefghijklhijklmno (21 chars) | Above Maximum | Error message displayed | Error displayed | Pass   |

### ***Email Field Boundary Testing***

| <b>Test Case ID</b> | <b>Input</b>   | <b>Boundary Type</b> | <b>Expected Result</b>  | <b>Actual Result</b> | <b>Status</b> |
|---------------------|----------------|----------------------|-------------------------|----------------------|---------------|
| SU-05               | usergmail.com  | Invalid Format       | Error message displayed | Error displayed      | Pass          |
| SU-06               | user@gmail.com | Valid Input          | Accepted                | Accepted             | Pass          |
| SU-07               | @gmail.com     | Missing Username     | Error message displayed | Error displayed      | Pass          |
| SU-08               | user@          | Missing Domain       | Error message displayed | Error displayed      | Pass          |

### ***Password Field Boundary Testing***

| <b>Test Case ID</b> | <b>Input</b>                 | <b>Boundary Type</b> | <b>Expected Result</b>  | <b>Actual Result</b> | <b>Status</b> |
|---------------------|------------------------------|----------------------|-------------------------|----------------------|---------------|
| SU-09               | abc1234 (7 chars)            | Below Minimum        | Error message displayed | Error displayed      | Pass          |
| SU-10               | abc12345 (8 chars)           | Minimum              | Accepted                | Accepted             | Pass          |
| SU-11               | abcd1234efgh5678 (16 chars)  | Maximum              | Accepted                | Accepted             | Pass          |
| SU-12               | abcd1234efgh56789 (17 chars) | Above Maximum        | Error message displayed | Error displayed      | Pass          |

### ***Phone Number Field Boundary Testing***

| <b>Test Case ID</b> | <b>Input</b>             | <b>Boundary Type</b>  | <b>Expected Result</b>  | <b>Actual Result</b> | <b>Status</b> |
|---------------------|--------------------------|-----------------------|-------------------------|----------------------|---------------|
| SU-13               | 0300123456 (10 digits)   | Below Required Length | Error message displayed | Error displayed      | Pass          |
| SU-14               | 03001234567 (11 digits)  | Exact Length          | Accepted                | Accepted             | Pass          |
| SU-15               | 030012345678 (12 digits) | Above Required Length | Error message displayed | Error displayed      | Pass          |
| SU-16               | 03abc234567              | Invalid Characters    | Error message displayed | Error displayed      | Pass          |

### Login Interface Testing

The Login module validates user credentials before granting access.

#### Fields Selected for Testing

| Field    | Validation Rule                |
|----------|--------------------------------|
| Email    | Must follow valid email format |
| Password | Must match registered password |

#### Test Cases for Login

##### Email Validation Testing

| Test Case ID | Input          | Boundary Type  | Expected Result           | Actual Result   | Status |
|--------------|----------------|----------------|---------------------------|-----------------|--------|
| LG-01        | usergmail.com  | Invalid Format | Error message displayed   | Error displayed | Pass   |
| LG-02        | user@gmail.com | Valid Format   | Proceed to authentication | Proceeded       | Pass   |
| LG-03        | empty field    | Null Value     | Required field error      | Error displayed | Pass   |

##### Password Validation Testing

| Test Case ID | Input            | Boundary Type | Expected Result       | Actual Result   | Status |
|--------------|------------------|---------------|-----------------------|-----------------|--------|
| LG-04        | empty field      | Null Value    | Required field error  | Error displayed | Pass   |
| LG-05        | Wrong password   | Invalid Input | Authentication denied | Access denied   | Pass   |
| LG-06        | Correct password | Valid Input   | Login successful      | Logged in       | Pass   |

## ***Test Execution Summary***

### **Module Total Test Cases Passed Failed**

|              |           |           |          |
|--------------|-----------|-----------|----------|
| Sign Up      | 16        | 16        | 0        |
| Login        | 6         | 6         | 0        |
| <b>Total</b> | <b>22</b> | <b>22</b> | <b>0</b> |

## **Results and Analysis**

The boundary value analysis testing verified that all input fields in the Sign Up and Login modules correctly enforce validation constraints. The system successfully rejected invalid inputs at boundary limits and accepted valid inputs within defined ranges.

The inclusion of phone number validation ensured that only correctly formatted 11-digit numbers are accepted, improving data consistency and system reliability.

Overall, the testing confirms that:

- Input validation is robust and reliable
- Boundary conditions are properly handled
- User authentication is secure
- The system prevents invalid or malformed data entry

## Work Division between group members

The RetailIQ project was developed collaboratively by two team members, with responsibilities clearly divided based on skill sets to ensure efficient development and timely completion of all project components.

### Team Members

- Ayesha Khalid
- Uma E Rubab

### Task Distribution

#### Ayesha Khalid (Backend & Machine Learning Development)

Ayesha was responsible for all backend development and machine learning components of the system. Her contributions include:

- Performing **Exploratory Data Analysis (EDA)** on the retail dataset
- Designing and implementing **data preprocessing and feature engineering**
- Developing and training **machine learning models for sales forecasting**
- Optimizing model performance and validating results
- Designing and implementing **backend APIs** for all system functionalities
- Handling **server-side logic and data processing**
- Testing APIs using **Postman** to ensure correct functionality
- Preparing **technical documentation** for different project phases

#### Uma E Rubab (Frontend Development & System Integration)

Rubab was responsible for designing and developing the user interface and integrating the system components. Her contributions include:

- Designing and developing the **frontend interface using modern UI frameworks**
- Creating all system pages including:
  - Login and Signup
  - Dashboard
  - Sales and Analytics pages
  - Recommendations page
  - Feedback and Announcements pages
  - Help/User Guide
- Implementing **interactive visualizations and charts**
- Ensuring **responsive and user-friendly design**
- Integrating **frontend with backend APIs**
- Managing **data flow between client and server**
- Testing overall system functionality from the user perspective

## Lesson learnt by group

Throughout the development of the RetailIQ project, the team gained valuable technical and collaborative experience. One of the key lessons learned was the importance of **effective communication, understanding, and adaptability within a team environment**.

During the initial phases of the project, most of the backend development and machine learning tasks were completed independently due to unforeseen circumstances affecting one team member's availability. However, once the situation improved, responsibilities were effectively balanced, with the remaining work—particularly frontend development and system integration—being successfully completed by other team member.

This experience highlighted that **project success is not solely dependent on equal distribution of work at all times, but rather on mutual support, trust, and flexibility among team members**. By maintaining a cooperative and understanding approach, the team was able to overcome challenges and ensure the timely completion of the project.

Additionally, the team learned the importance of:

- Proper task division based on individual strengths
- Maintaining clear documentation for smooth handovers
- Ensing seamless integration between different system components
- Staying adaptable in dynamic project situations

Overall, the project reinforced that a **positive and supportive team dynamic** plays a critical role in successfully delivering complex software systems.